



Active Learning Modules For Training Engineers To Develop Innovative Orthopedics Applications

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Acknowledgements

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Teaching resources modified from:

Andrew Gerhart, PhD

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Lawrence Tech

Kettering University

Additional teaching resources can be found at:

<https://engineeringunleashed.com/cards/cardview.aspx?CardGuid=6513d2b9-0ed7-4e45-a6d7-9119b9c663f0> (requires free registration)



Active Learning Modules to Simulate Surgery in "Engineering Applications in Orthopedics"

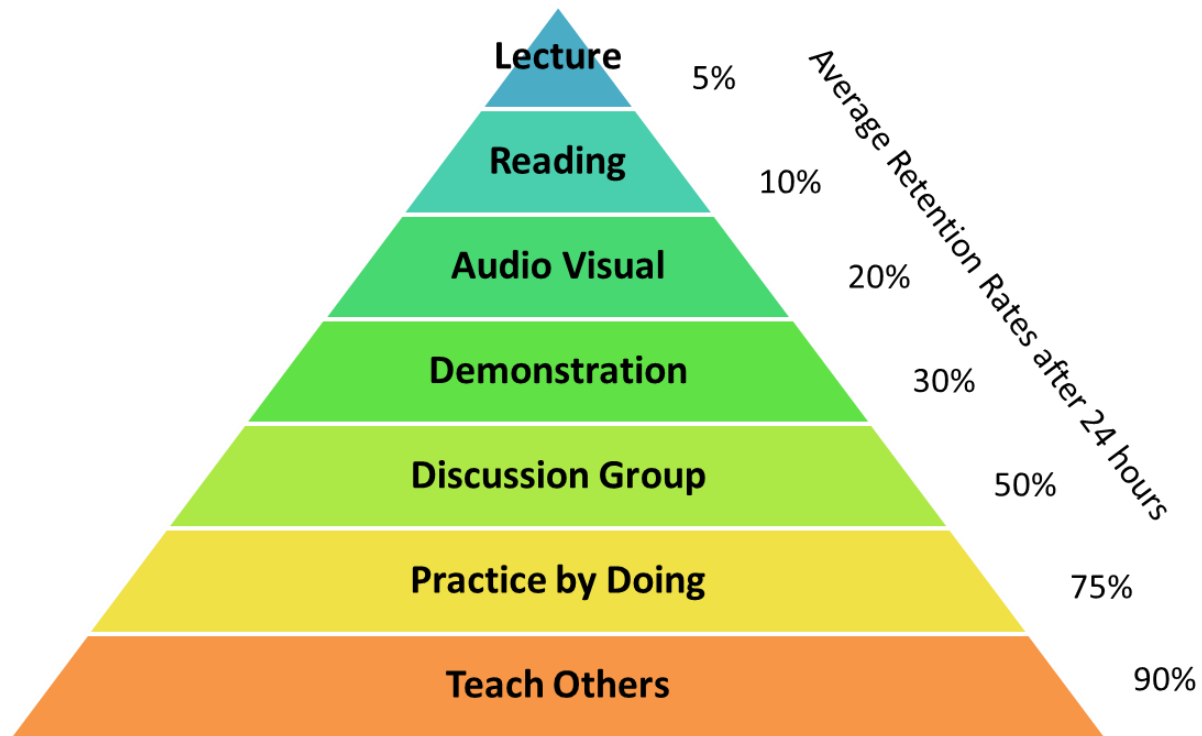
February 04, 2020 | by Eric Meyer

Implementation of 5 orthopedic surgery simulations to help engineers communicate with clinicians, recognize opportunities, and improve medical devices

[Read More ▶](#)

Learn by Doing

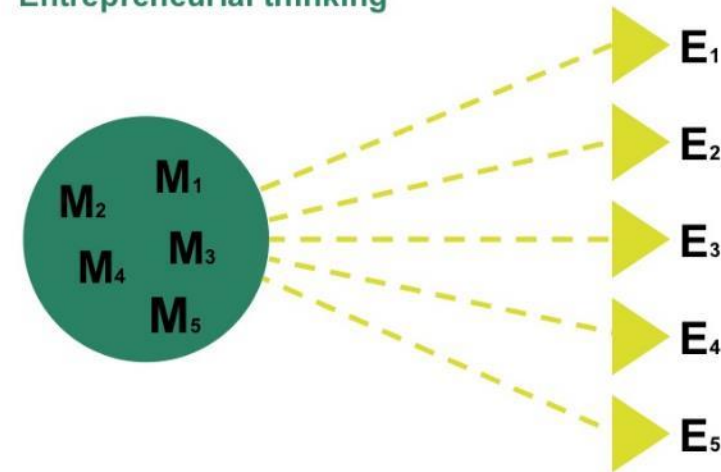
The Learning Pyramid



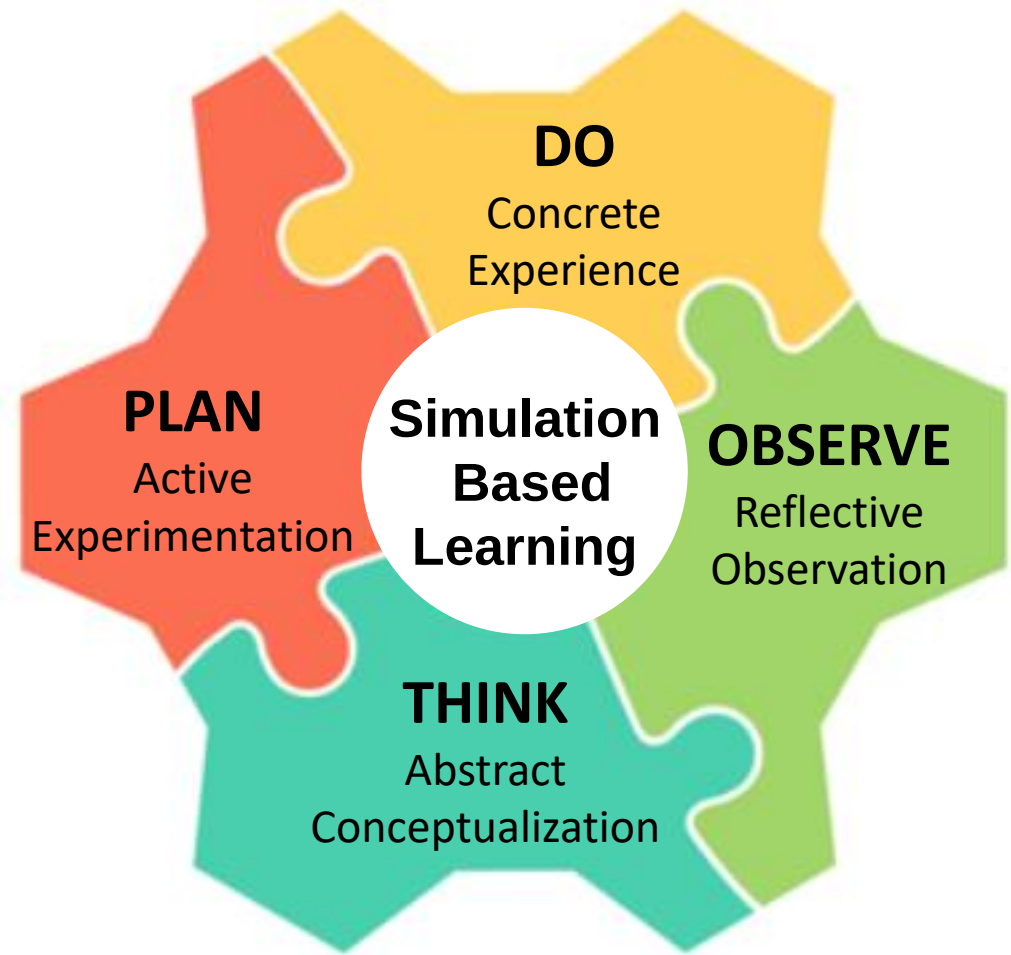
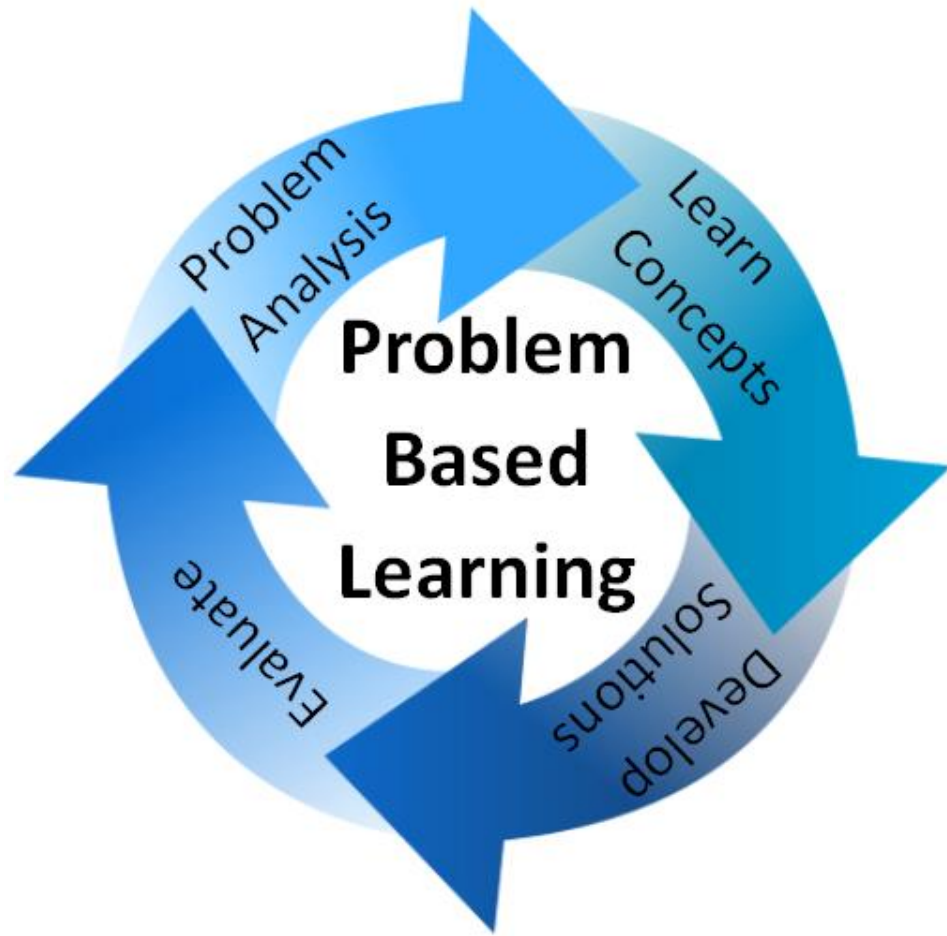
Source: National Training Laboratories, Bethel Maine



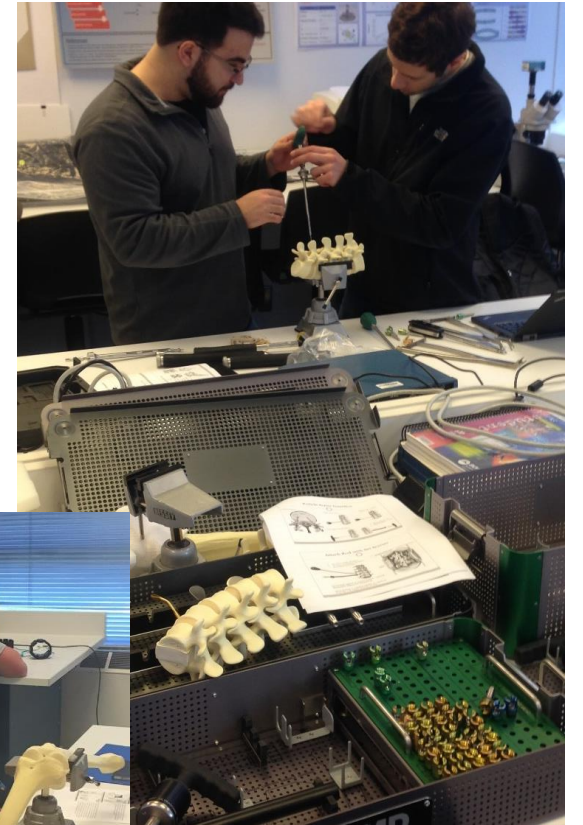
Effectual reasoning Entrepreneurial thinking



Pedagogy



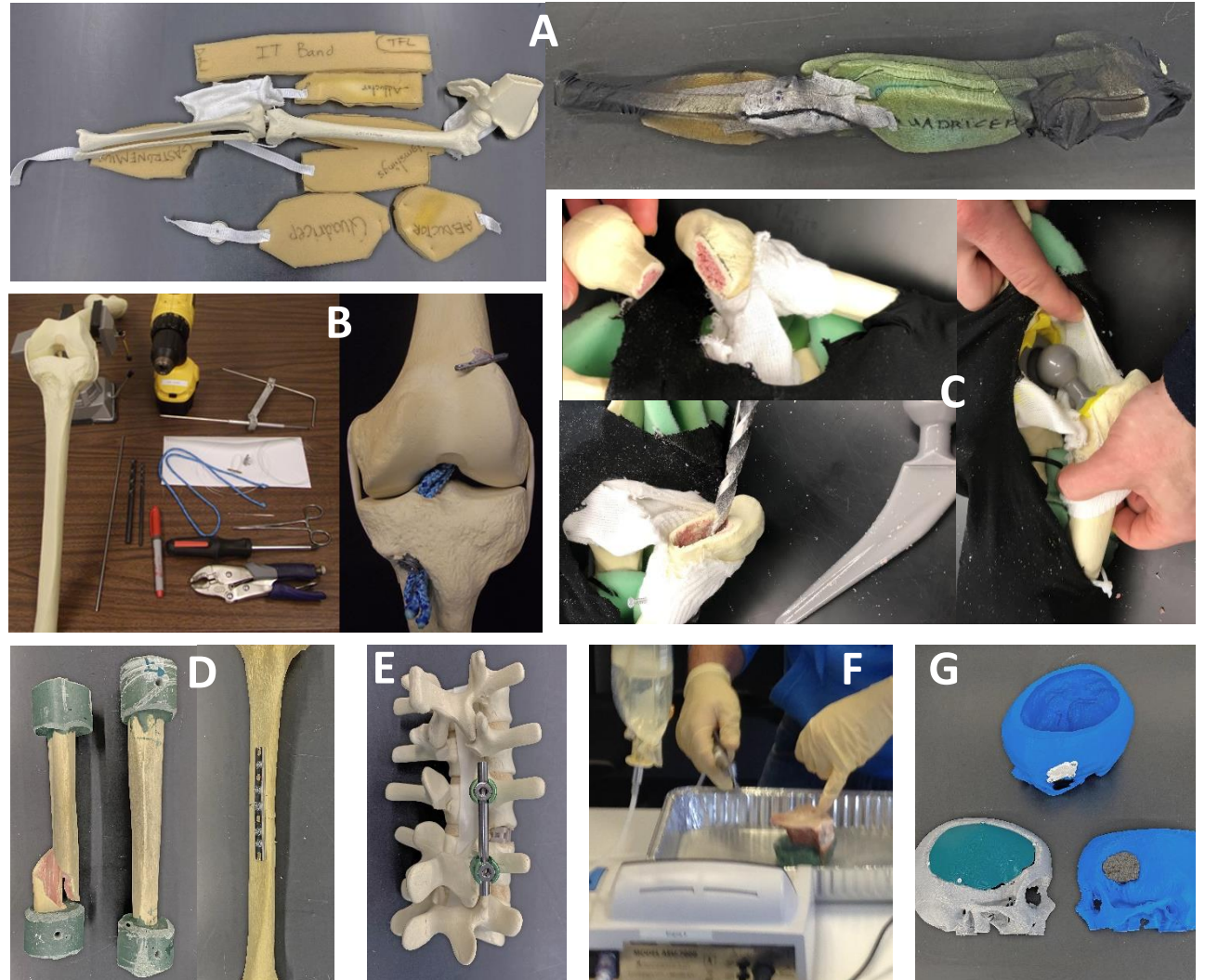
Simulation Methods



BME 5313

Engineering Applications in Orthopedics

Week	Topic	In Class Activity
1	Introduction	Virtual Hip Surgery
2	Sports Injuries	A Build Lower Extremity
3	Ligament Tear	B ACL Reconstruction
4	Cartilage/Arthritis	C Hip Replacement
5	Bone Fracture	D Fracture Fix
6	Spinal	E Spine Fixation
7	Skeletal Development	Material Testing
8	Implant Stability	F Dental Implants
9	Custom Implants	G Mimics Skull Plate
10	Medical Device Development	Opportunity Creation
11	Biomech Modeling	Seminar
12	Prosthetics/ Orthotics	Industry Tour
13	Injury Prevention	Clinic Tour
14	Medical Imaging	Hospital Tour
15	Concussion	Seminar



Broader Projects

Weekly Journal Club

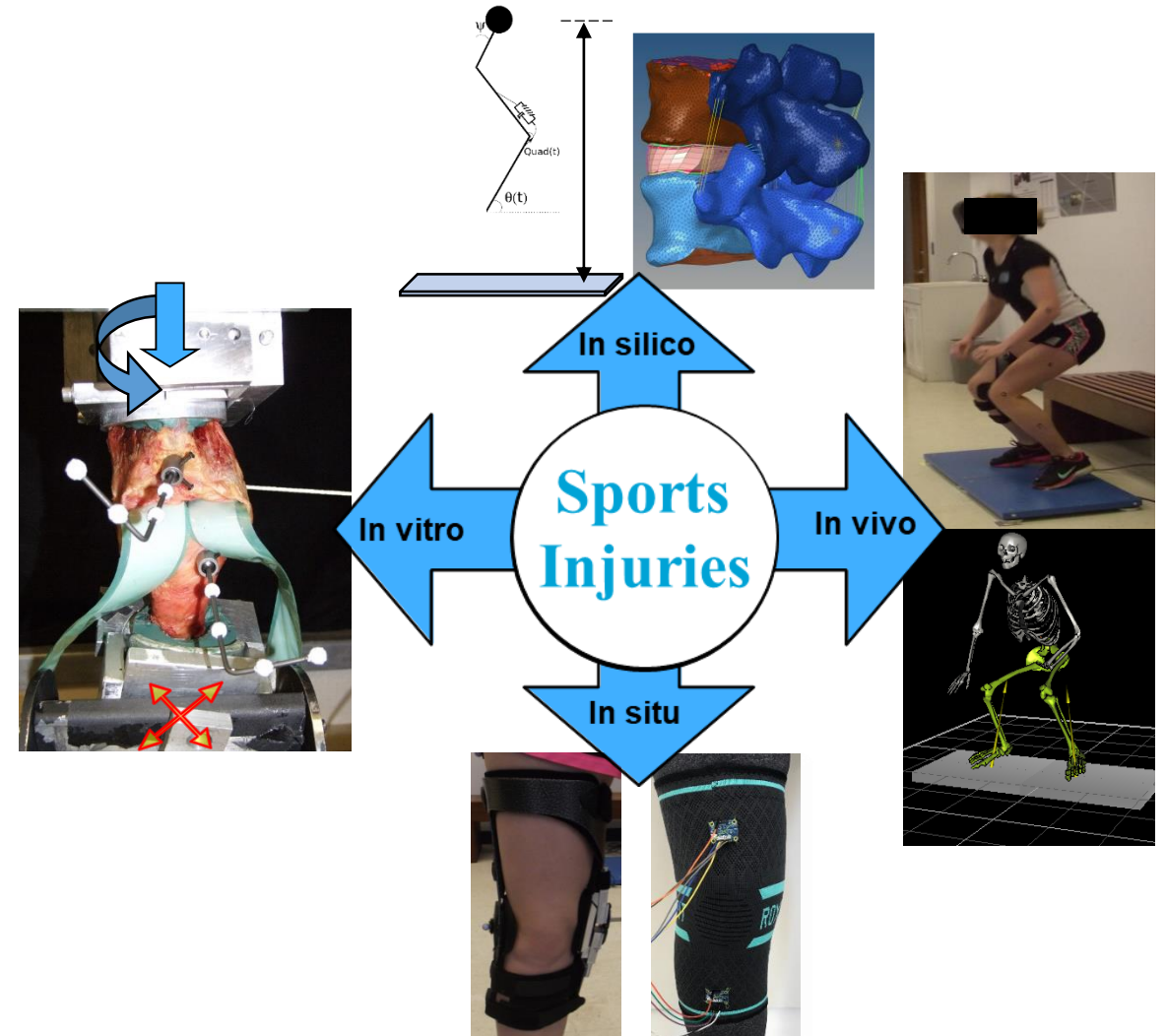
Peer-reviewed Research Paper

Opportunity Creation Project

Medical Devices Regulation

Orthopedics Industry

Clinical/Hospital Tours



Engineers + Clinicians



A surgeon has contacted you through the LTU Innovation Center because of your expertise in Biomedical Engineering.

PBL: Build and Test a new type of Orthopedic Implant

- Understand clinical treatment procedures
- Device Concepts
- Prototype Fabrication
- Testing and Validation
- Business Model and Usability
- IP and FDA documentation
- Research report and clinical meeting presentations

